

PRAHARI

Predictive Routing & AI for Hotels, Attractions, Regions & Itineraries

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AI Demand Forecasting & Crowd Rebalancing to Boost Hotels, Homestays and Travel in India

PREDICT · PROTECT · REDISTRIBUTE

PRAHARI forecasts where and when tourists will go, compares that forecast with each destination's live carrying capacity, and redirects surplus demand to nearby "twin" destinations and small hotels that have room. It is built on ONDC, Bhashini and UPI, so overcrowding becomes income for local tourism businesses instead of a crisis.

7–14 days

early warning of peak crowds

+10–15 pts

off-season occupancy for small stays

< 10 min

host onboarding on WhatsApp

Single-digit

fees vs 15–25% commission

Pilot targets, to be validated in the field.

Problem Statement

Student Innovation: a solution that can boost the tourism industry, including hotels, travel and others

Document type

Additional Document: Business Model Canvas, Technical Architecture & Impact Assessment

Target users

State & district tourism bodies · Hotel, homestay & travel MSMEs · Domestic & foreign travellers

Built on

ONDC · Bhashini · UPI · NIDHI · DigiLocker · NDMA SACHET · 112 ERSS · 1363 helpline

1 • The problem is **mismatch**, not shortage

Too many tourists in too few places on too few days, while nearby hotels and homestays stay empty.

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Problem & Solution

2,509 M

Domestic tourist visits in 2023
(India Tourism Statistics 2024)

9.52 M

Foreign tourist arrivals in 2023,
still below 10.93 M in 2019

~5%

Tourism's share of India's GDP
(Tourism Satellite Account)

15–25%

Typical OTA commission paid by
small hotels & homestays

Reactive

Crowd control today: e-pass caps
(Ooty 2024), NGT permits
(Rohtang)

Five pain points we solve

- **Overtourism at hotspots:** hours-long traffic jams, water and waste stress, crowd-safety risk on long weekends and festivals.
- **Empty rooms nearby:** MSME stays sit half-empty for most of the year with strong seasonality.
- **Digital exclusion:** small hosts are unlisted or dependent on high-commission OTAs, and face language barriers.
- **No foresight for officials:** tourist data arrives months late, so there is no forecast for next weekend.
- **Trust & safety gaps:** overcharging, fake listings, and tourists stranded by monsoon disasters.

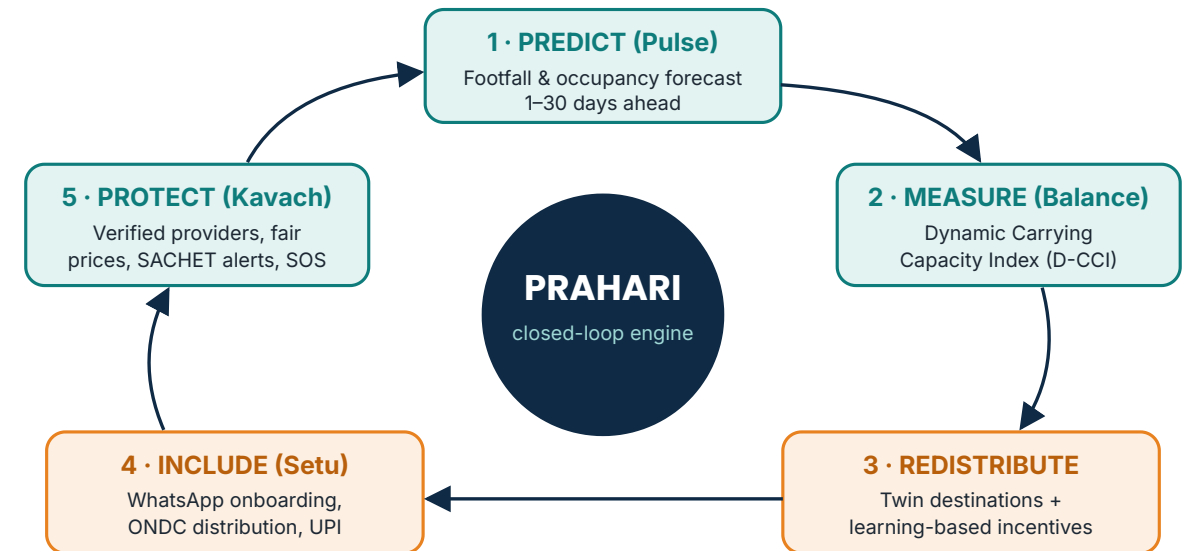
Why existing solutions fail

- OTAs **sell** rooms • Portals **inform** • Permits **restrict** • Map apps **react**
- No one connects demand forecasts with carrying capacity and redistribution.

Who is affected

Travellers	Lost time, surge prices, safety risk, poor experience
Hotels & MSMEs	Off-season losses, commission drain, no demand visibility
Communities	Congestion, water & waste stress, rising resentment (Doxey's Irridex)
Governments	Rescue costs, lost revenue, declining destination brands (Butler's life cycle)

PRAHARI closes the loop



Drishti dashboard gives governments visibility and control over every step of the loop

GREEN load < 0.8

AMBER 0.8 – 1.0

RED > 1.0 → rebalance

BLACK hazard / closed

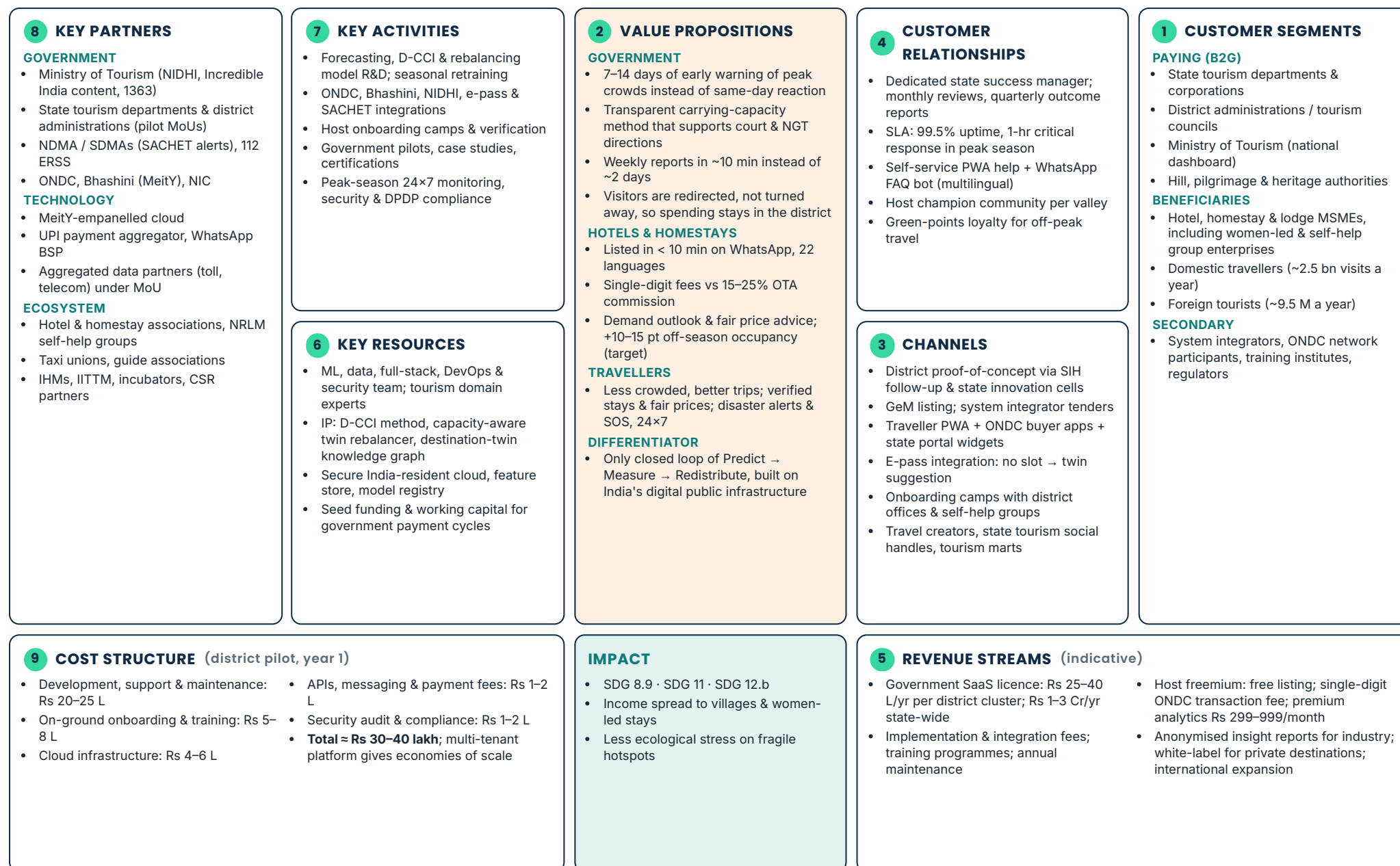
Example in 30 seconds: Manali is forecast **RED** for the long weekend (2.1× normal crowd) → PRAHARI suggests Tirthan Valley (**GREEN**, 3 h closer, 38 verified homestays, 15% off) → the traveller books via ONDC + UPI → homestay owner Ramesh gets a booking he would never have received → the Kullu District Tourism Officer saw the peak a week earlier on Drishti and pre-positioned police, buses and sanitation.

2 • Business Model Canvas

Commercial viability, social impact and scalability. All financial figures are indicative estimates for a district pilot.

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Business Model

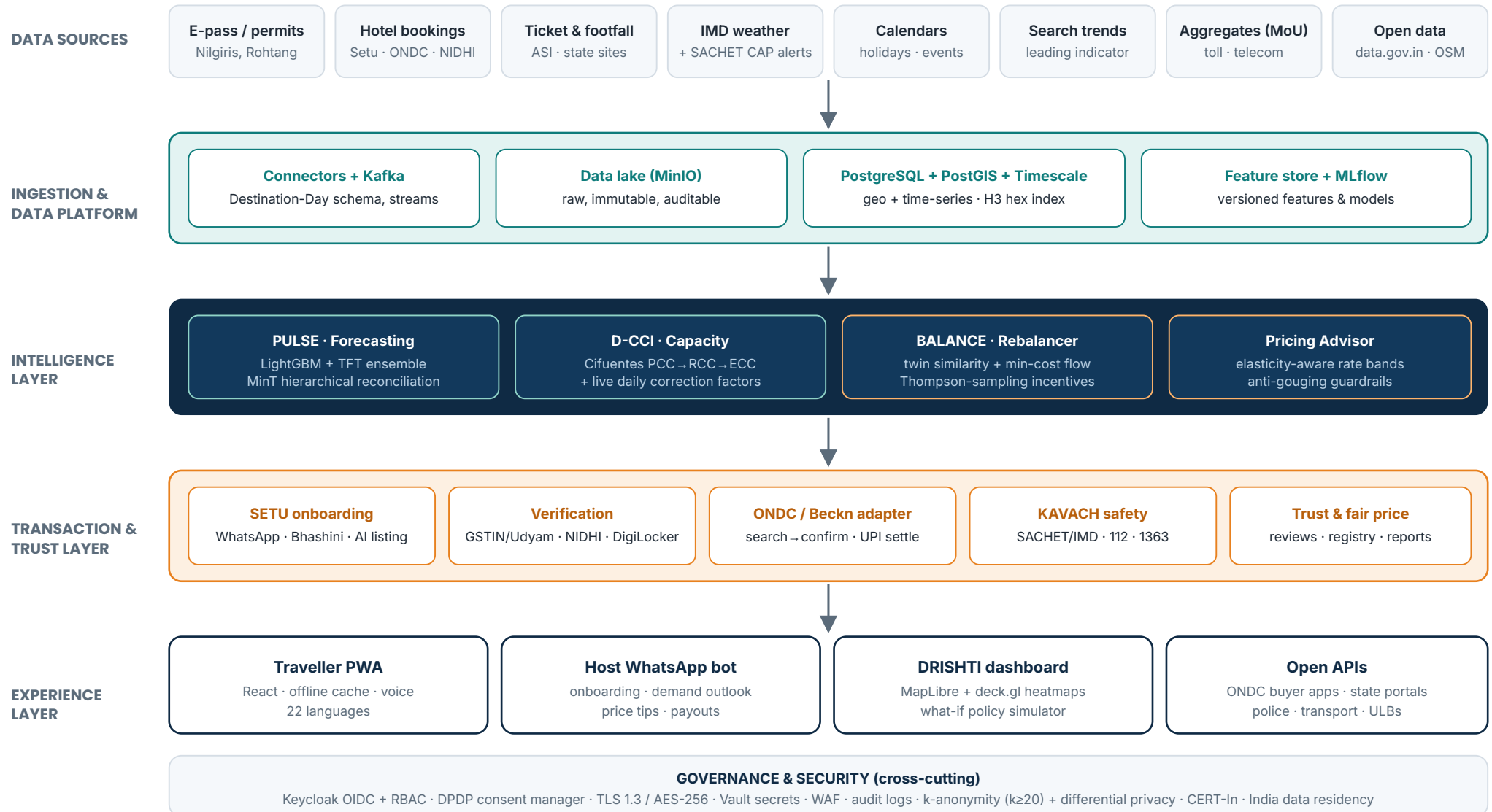


3 • System Architecture

Layered, event-driven microservices on an India-resident cloud, with open standards throughout (Beckn/ONDC, OpenAPI, CAP).

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Technical Documentation



4 • Core Algorithms & Technology Choices

Input → Processing → Output: proven techniques combined in a new way, with every tool justified.

A. PULSE: multi-horizon demand forecasting

- **Features:** lags (1, 7, 14, 28, 364 days), rolling statistics, long-weekend flags, school-vacation overlap, festival & event intensity, weather anomalies, search momentum, origin-city demand.
- **Models:** LightGBM (fast, SHAP-explainable) + Temporal Fusion Transformer (known-future inputs), with ETS/ARIMA fallbacks for sparse series.
- **Consistency:** MinT reconciliation across attraction → destination → district → state. **Cold start:** transfer from a cluster of similar destinations.

B. D-CCI: dynamic carrying capacity & status

$PCC = (A / a) \times R_f$ // area per visitor × daily rotations
 $RCC = PCC \times \Pi(1 - C_{fi})$ // weather, landslide, roads, parking, water
 $ECC = RCC \times MC$ // management capacity (staff, services)
 $D-CCI(d,t) = ECC$ with C_{fi} recomputed daily
 $LR = Forecast(d,t) / D-CCI(d,t) \rightarrow G < 0.8 \cdot A \ 0.8-1 \cdot R > 1 \cdot Black = hazard$

Based on Cifuentes (1992), extended from a one-time study into a live daily index.

C. BALANCE: capacity-constrained twin rebalancing

- **Twin candidates:** cosine similarity of attribute and description embeddings, with a travel-time penalty and spare-capacity filter.
- **Allocation:** min-cost flow (OR-Tools) sends overflow nudges without pushing any twin above **0.8 × its own D-CCI**, which prevents overflow cascades.
- **Incentives:** a contextual Thompson-sampling bandit learns which nudge (discount, free experience, priority slot, green points) works for which segment, within budget.
- **Fairness:** exposure is rotated among eligible small stays so large properties are not favoured.

Layer	Choice	Why
Traveller app	React PWA (Vite, TS)	No install; works on low-end phones; offline cache in the mountains
Host channel	WhatsApp Cloud API	Hosts already use it daily; zero learning curve
Languages	Bhashini ASR/MT/TTS	22 scheduled languages; government platform
APIs	FastAPI (Python)	Async, OpenAPI docs, same language as the ML stack
Commerce	Beckn/ONDC adapter	Open network; single-digit fees; reaches all buyer apps
Streaming	Apache Kafka	Decouples permits and bookings from real-time consumers
Storage	PostGIS + TimescaleDB	Geo queries + fast time-series aggregation, open source
Spatial	Uber H3	Uniform hexagons for heatmaps and privacy thresholds
ML	LightGBM, PyTorch-Forecasting, OR-Tools	Accuracy + interpretability + optimisation
MLOps	MLflow, Evidently	Reproducible models; drift alerts each season
Dashboard	MapLibre + deck.gl	No map licence fees; renders thousands of hexagons
Infra	Docker, Kubernetes, MeitY cloud	Scales for peak season; India data residency

Privacy & responsible AI by design

- No individual tracking. Crowd data is published only at H3 level with $k \geq 20$ and differential-privacy noise.
- DPDP Act 2023 consent, purpose limitation and erasure. SHAP explanations. Humans approve all policy actions.

5 • Market & Competitive Landscape

Why PRAHARI is more than 30% better on the dimensions that matter to this Problem Statement.

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Market Analysis

Capability	OTAs (MakeMyTrip, Booking, Airbnb)	Govt. tourism portals	E-pass / permit systems	Map & traffic apps	AI trip planners / chatbots	PRAHARI
Forecasts demand days ahead	✗	✗	✗	Same-day	✗	✓ 1-30 days
Knows carrying capacity	✗	✗	Fixed cap	✗	✗	✓ Dynamic D-CCI
Redirects surplus demand	✗	✗	✗ turns away	Re-routes cars	Unconstrained	✓ Capacity-aware twins
Includes unlisted MSME stays	15-25% fee	✗	✗	✗	✗	✓ WhatsApp + ONDC
Local-language, voice-first	Partial	Partial	Partial	Partial	Partial	✓ 22 languages
Safety: alerts, SOS, fair price	✗	Helpline	✗	Traffic only	✗	✓ SACHET, 112, 1363
Policy tools for government	✗	✗	Quota only	✗	✗	✓ What-if simulator

Market opportunity

- **B2G:** 36 states and UTs, 780+ districts; 57 Swadesh Darshan 2.0 destinations; 40 SASCI iconic-centre projects; the Budget 2025-26 plan to develop 50 top destinations.
- **Hosts:** the large, unorganised base of small hotels, lodges and homestays, with branded inventory under 2 lakh rooms. Homestays have MUDRA-loan support in Budget 2025-26.
- **Travellers:** ~2.5 billion domestic visits and ~9.5 million foreign arrivals a year.

Beachhead market

- Hill-station and pilgrimage clusters where courts and regulators already require capacity controls: Kullu-Manali, the Nilgiris (Ooty, Kodaikanal), Mussoorie, Darjeeling and the Char Dham routes.
- These buyers already feel the pain and have budgets, so they are the fastest path to a paid pilot.

Target-user research plan

- **Hosts (n ≥ 50):** occupancy by month, OTA use and commission, WhatsApp usage, language, willingness to pay.
- **Travellers (n ≥ 200):** would you switch to a quieter twin? Which incentive works? What is your trust threshold?
- **Officials (5-8 interviews):** DTOs and state tourism IT on data gaps, e-pass pain points and procurement route.
- Results feed the incentive-bandit priors and UX decisions before the pilot.

Validation metrics

- Share of travellers who accept a twin suggestion (target ≥ 10%).
- Host activation: listing completed within 10 minutes (target ≥ 70%).
- Officials: weekly dashboard use and forecast-versus-actual accuracy.

Defensible advantages

- D-CCI scoring method and capacity-aware rebalancer (patent potential).
- Destination-twin knowledge graph for India.
- Deep integration with ONDC, Bhashini, NIDHI, SACHET, 112 and 1363.
- **Data network effect:** every host and every season improves forecasts and recommendations.
- Open standards, so states adopt without lock-in, which makes PRAHARI a candidate for tourism Digital Public Infrastructure.

Why now

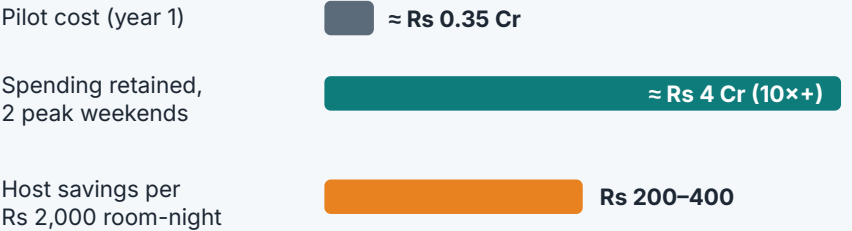
- ONDC, Bhashini and UPI are mature enough to build on.
- Budget 2025-26 destination development and SASCI investments need outcome tracking.
- Climate extremes make early warning a necessity.

6 · Impact, Cost-Benefit & Roadmap

Pilot KPIs are targets to be validated in the field; financials are indicative estimates.

Metric	Today	With PRAHARI (pilot target)
Advance warning of peak crowd	Same day / reactive	7–14 days
7-day footfall forecast error	No forecast	MAPE ≤ 15%
Peak overflow redirected to twins	0% (turned away)	10–15%
Off-season occupancy of onboarded MSME stays	Baseline	+10–15 percentage points
Distribution cost for hosts	15–25% OTA commission	Single-digit % via ONDC
Host onboarding time	Days	< 10 minutes (WhatsApp)
Disaster alert reach to tourists	Ad hoc	< 5 minutes from SACHET alert
Weekly destination report	~2 days manual	~10 minutes automated

ROI for a district pilot (illustrative)



Assumption: 1 lakh peak-weekend visitors × Rs 4,000 spend = Rs 40 Cr at stake. Retaining 5% gives ≈ Rs 2 Cr per weekend. Excludes GST, off-season gains and avoided rescue costs.

Scaling roadmap: prototype to national Digital Public Infrastructure

HACKATHON · 36 h

Working prototype

Kullu-Manali + 4 twins; PWA, WhatsApp bot, forecasting, rebalancer, dashboard; public + simulated data.

MONTHS 0–9

District pilot

MoU with Kullu / Nilgiris; ONDC certification; 300–500 MSME stays; measured KPIs and a published case study.

MONTHS 9–18

State rollout

3–5 states as SaaS tenants; e-pass and state portal integration; guides, taxis and experiences added.

MONTHS 18–36

National scale

All priority destinations; industry APIs; tourism Digital Public Infrastructure; South Asia and ASEAN expansion.

SDG 8.9

Sustainable tourism that creates local jobs

SDG 11

Safer, less congested towns

SDG 12.b

Tools to monitor the impact of sustainable tourism

Viksit Bharat

Dekho Apna Desh · Swadesh Darshan 2.0 · Digital India

Stakeholder	Economic benefit	Social / experience benefit	Long-term benefit
Travellers	Incentives on twins; fair, transparent prices	Less crowding, safer trips, local-language help	Discovery of new destinations; loyalty rewards
Hotels & homestays	Higher occupancy; lower commission; better pricing	Digital inclusion for rural & women-led stays	Stable year-round income; access to MUDRA credit through booking history
Governments	Retained spending & GST; lower rescue costs	Higher visitor & resident satisfaction	Evidence-based planning of infrastructure and schemes
Environment & communities	Income spread across villages	Less congestion, water & waste stress	Longer destination life cycle; protected fragile ecosystems

7 • 36-Hour Build Plan, Risks & References

A realistic scope, pre-built foundations and offline fallbacks make the demo robust.

0-4 h Scope lock (Kullu-Manali + 4 twins); load datasets; repo, CI, Docker Compose	4-12 h Pulse pipeline (LightGBM + TFT), D-CCI calculator, core APIs; PWA screens; dashboard layout	12-20 h Twin similarity + min-cost flow; WhatsApp bot; Bhashini; ONDC sandbox flow	20-28 h Kavach alerts + SOS mock; pricing advisor; what-if simulator; end-to-end tests	28-32 h Auth + RBAC hardening; load tests; MAPE on hold-out; bug fixes	32-36 h Demo polish; backup video; pitch rehearsal; Q&A sheet; documentation
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Team of 6 & key risks

Roles: ML lead · Data engineer · Backend & integration · Frontend/PWA · Dashboard & UX · Domain research, testing & pitch

Risk	Mitigation
Sparse footfall data	Proxy signals, cold-start transfer, state data MoUs
Host adoption	WhatsApp and language-first design, zero joining fee, camps with self-help groups
Travellers ignore nudges	Learned incentives, honest delay estimates, e-pass twin offers
Overflow cascade to villages	Hard cap at 0.8 × twin D-CCI; community consent
API dependency	Modular adapters, cached fallbacks, early ONDC and Ministry of Tourism engagement
Privacy	Aggregate-only data, DPDP consent, independent audits

Done before the finale (de-risking)

- Cleaned datasets and feature pipeline; LightGBM baseline with hold-out MAPE.
- Twin-destination attribute sheet for the pilot cluster; UI component library.
- Sandbox access tested (ONDC, Bhashini, WhatsApp, UPI) with cached fallback responses.
- Backup demo video recorded, so a failed network cannot fail the demo.

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